COMPREHENSIVE PROJECT & PRESENTATION DOSSIER

Sehat Sakha (सेहत सखा)

"Aapki Report, Aapki Bhasha Mein" (आपकी रिपोर्ट, आपकी भाषा में)

An AI-Powered, Rural-First, Bilingual Medical Laboratory Report Simplifier, Visual Analytics Platform & Native Hindi Voice Companion

LIVE PRODUCTION WEBSITE
https://ai-medical-simplifier.vercel.app

AI MASCOT & COMPANION
Sakha (सखा — Trusted Health Friend)

CORE STACK
React 19, TypeScript, Vite, Google Gemini Multimodal AI, Vercel Edge TTS

> 📌 **Guide for AI / ChatGPT Presentation & Script Generation**
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> This document is the complete, authoritative specification of the **Sehat Sakha** web application. It covers the problem statement, rural UX philosophy, end-to-end user journey, 12 core features, hybrid AI + deterministic architecture, native Hindi Text-to-Speech engineering, visual analytics (Mode B), and presentation script talking points. Use this report to generate slide decks, pitch scripts, live demo walkthroughs, and Q&A responses.

1. Executive Summary & Problem Statement

Across India—especially in rural and semi-urban areas—millions of patients receive diagnostic laboratory reports (such as Complete Blood Count, Lipid Profile, Liver/Kidney Function Tests, Thyroid Panels, and Blood Sugar tests) printed exclusively in complex English medical jargon. When patients or elderly family members look at terms like "*Total Leukocyte Count*", "*Serum Creatinine*", "*Erythrocyte Sedimentation Rate*", or raw numeric intervals like `13.5 - 17.5 g/dL`, they experience anxiety, confusion, and helplessness.

⚠️ The Real-World Healthcare Gap

1. **Language Barrier:** Lab reports are printed in English, whereas over 70% of rural and elderly patients prefer Hindi or regional languages.
2. **Numeracy & Graph Barrier:** Standard clinical tables fail to visually communicate whether a value is slightly off or critically abnormal.
3. **Illiteracy & Low Vision:** Elderly patients struggle to read dense text on budget mobile screens and need **spoken audio** in their native tongue.
4. **Rushed Doctor Visits:** Patients often forget what to ask the doctor during short 3-minute consultations.

💡 The Sehat Sakha Solution

Sehat Sakha ("Health Friend") bridges this gap by combining **Google Gemini Multimodal Vision OCR**, a **Deterministic Clinical Reference Engine**, **Native Hindi Voice Synthesis**, and **Intuitive Visual Metaphors (Building Graphs, Pie Charts, 3-Zone Range Meters)** into an ultra-accessible web app designed specifically for entry-level Android phones and users with limited digital literacy.

2. Product Vision & Design Philosophy

- **"Simple by Default, Detailed When Requested" (Progressive Disclosure):** A first-time user is never overwhelmed by complex charts on the initial screen. Mode A shows plain-language explanations and a 1-tap Hindi Audio Player first; Mode B unlocks rich interactive visualizations (Pie Chart, Building Graph, Range Meters, Trend Lines) on demand.
- **Strict Clinical Safety & Non-Diagnostic Guardrails:** Sehat Sakha is an educational health literacy tool, **never a replacement for a doctor**. It strictly avoids diagnosing diseases or prescribing medicines, instead empowering patients with plain-language physiological context and a printable list of *Questions to Ask Your Doctor*.
- **100% Real Data Integrity (Zero Mock/Fake Data in Production):** Unlike demo prototypes that display hardcoded patient profiles, Sehat Sakha starts with an honest empty state ("*Abhi koi report save nahi hai*") and extracts 100% real data from the user's uploaded report image or PDF.
- **Rural-First Accessibility (Entry-Level Android Optimization):** Built with high-contrast colors (`#064e3b` forest emerald), minimum `48px–54px` touch targets, explicit text labels alongside every icon, and bilingual Hindi/English UI everywhere.

3. End-to-End 5-Stage User Workflow

STAGE 1 **Bilingual Onboarding & Multi-Format Report Upload**

When a first-time user opens Sehat Sakha, a friendly bilingual modal (`LanguageSelectorModal.tsx`) lets them choose **हिंदी (Hindi)** or **English** with large touch cards. On the home screen, the user taps the prominent **"रिपोर्ट अपलोड करें (Upload Report)"** button to take a camera photo or select an image (JPG, PNG, WebP), PDF document, or text file of their lab report.

STAGE 2 **Real Gemini Multimodal OCR & Human-in-the-Loop Extraction Review**

Before generating explanations, Sehat Sakha invokes `extractMedicalParametersFromFiles()` using Google Gemini Multimodal Vision models (`gemini-3.1-flash-lite` / `gemini-3.5-flash-lite`). It reads every biomarker name, Hindi translation, patient value, unit, and printed reference interval from the image/PDF. The user is then shown the **Extraction Review Screen (`ExtractionReview.tsx`)** where they can verify the extracted numbers against their paper report or manually edit any value if their camera photo was blurry. Editing a value marks its badge as "*Manual / आपके द्वारा संशोधित*" and immediately recalculates its clinical status.

STAGE 3 Mode A — Simple Explanation & Native Hindi Voice Playback (Default View)

Once confirmed, the report is automatically saved to the user's **My Reports (मेरी रिपोर्ट्स)** vault with a Date & Time stamp, and the user lands on **Mode A (`SimpleExplanationView.tsx`)**. This view presents:

- **Top Overview Banner (`ReportOverviewBar.tsx`):** Shows patient details, lab name, total tests count, how many are within normal range (green), and how many need attention (amber/red).
- **1-Tap Hindi Audio Bar ("हिंदी में पूरी बात सुनें"):** Streams natural Hindi speech explaining the report sentence-by-sentence with live subtitle tracking, Pause, Resume, and Stop controls.
- **Parameter Summary Cards:** Plain-language cards for every biomarker with a "*सखा से सुनें (Listen to Sakha)*" button on each card.
- **5-Part AI Narrative Breakdown:** Generated by Gemini in the chosen language while preserving English test names so the patient can match them to their physical sheet.

STAGE 4 Mode B — Interactive Visual Analytics ("Detailed When Requested")

Tapping **"ग्राफ और चार्ट देखें (View Detailed Graphs)"** opens **Mode B (`DetailedAnalysisView.tsx`)**, which transforms abstract numbers into 4 visual metaphors designed for non-technical users: (1) **Health Pie Chart**, (2) **Building Bar Graph (इमारत ग्राफ)**, (3) **3-Zone Horizontal Range Meters**, and (4) **Historical Trend Graphs**. Every chart has a plain-Hindi text summary underneath and a button to ask Sakha to explain that specific chart.

STAGE 5 Ask Sakha Conversational Assistant, Doctor Questions & Auto-Saved Vault

At any point, the user can open the **Ask Sakha Drawer (`AskSakhaDrawer.tsx`)** to ask questions in Hindi or English (via quick-tap voice chips or typing) and listen to Sakha's spoken reply. They can also review the **Doctor Questions Checklist (`DoctorQuestions.tsx`)** before their clinic visit, or open the **Sidebar Drawer (`Sidebar.tsx`)** to access **My Reports (`HealthVault.tsx`)** where all past analyses are organized chronologically with date and time stamps.

4. Deep-Dive: 12 Key Features of Sehat Sakha

Feature / Module	Source File(s)	How It Works & Why It Matters for Presentation
1. Real Multimodal OCR Extraction	`src/lib/gemini.ts`	Uses Gemini multimodal vision to extract structured JSON containing `reportTitle`, `hindiTitle`, `patientName`, `labName`, `collectionDate`, and an array of all `LabParameter` items directly from camera photos or PDFs. Zero dummy data.
2. Deterministic Clinical Reference Engine	`src/lib/referenceEngine.ts`	AI never guesses whether a number is High, Low, or Normal. A dedicated regex & math engine (`parseRangeBounds` & `classifyValue`) parses two-sided (`13.5 - 17.5`) and one-sided (`< 200` , `> 50`) bounds to deterministically classify every value.
3. Human-in-the-Loop OCR Verification	`src/components/ExtractionReview.tsx`	Displays extracted test names, values, units, and reference ranges in an editable table before analysis. If a crumpled paper report caused an OCR misread, the user can fix the number in 1 tap.
4. Native Hindi Text-to-Speech (TTS) Engine	`src/lib/speech.ts` `api/tts.js`	Solves the notorious problem where mobile browsers lack local Hindi voices. Streams natural Google Neural Hindi audio via a custom Vercel Serverless proxy (`/api/tts`) with sentence chunking, live subtitles, and medical unit expansion.
5. Medical Unit Phonetic Expansion	`src/lib/speech.ts`	Automatically converts cryptic abbreviations into spoken Hindi before synthesis: e.g., `mg/dL` → *"मिलीग्राम प्रति डेसीलीटर"*, `g/dL` → *"ग्राम प्रति डेसीलीटर"*, `WBC` → *"सफेद रक्त कोशिकाएं"*, `Hb` → *"हीमोग्लोबिन"*.
6. Health Pie Chart (स्वास्थ्य पाई चार्ट)	`src/components/HealthPieChart.tsx`	Interactive SVG Donut/Pie chart showing the proportion of tests that are Normal (Green), High (Red), and Low (Amber), with a large center percentage score and plain-language summary card.

Feature / Module	Source File(s)	How It Works & Why It Matters for Presentation
7. Building Bar Graph (इमारत तुलना ग्राफ)	`src/components/BuildingBarGraph.tsx`	Uses a relatable rural metaphor: compares the patient's result ("आपकी इमारत / Your Building") side-by-side against the healthy maximum limit ("सामान्य सीमा इमारत") with a dashed green safe-zone roof line.
8. 3-Zone Range Meter (रेंज मीटर)	`src/components/RangeIndicator.tsx`	Horizontal gauge divided into 3 equal visual zones: Below Range (0–33%), Safe Normal Range (33–66%), and Above Range (66–100%). Includes text labels on every zone so colorblind users can read it effortlessly.
9. Longitudinal Trend Graph (ट्रेंड ग्राफ)	`src/components/TrendGraph.tsx`	Plots previous vs. current biomarker readings over time against a shaded green normal band. Never fabricates history—if only 1 report exists, it honestly instructs the user to upload a second report to unlock trends.
10. Contextual "Ask Sakha" Assistant	`src/components/AskSakhaDrawer.tsx`	Slide-over chat & voice companion pre-loaded with the user's exact biomarker context. Includes 1-tap question chips (e.g., "*इसमें कौन सी जांच सामान्य से अलग है?*") and instant Hindi voice readout.
11. Doctor Consultation Question Builder	`src/components/DoctorQuestions.tsx`	Generates targeted, non-alarmist questions in Hindi and English based specifically on out-of-range parameters, with interactive checkboxes so patients can track what they asked their doctor.
12. Auto-Archived Health Vault ("My Reports")	`src/components/HealthVault.tsx` `src/components/Sidebar.tsx`	Every analyzed report is automatically saved to local storage with exact Date (`12 Oct 2026`) and Time (`07:45 PM`) badges. Kept cleanly inside the Sidebar Drawer so the home screen stays clutter-free.

5. Technical Architecture & Engineering Highlights

5.1 Multi-Model Gemini AI Resilience Cascade

To prevent API rate-limit failures or regional model deprecations during live demonstrations, `src/lib/gemini.ts` implements an automatic fallback cascade across 7 multimodal models:

`gemini-3.1-flash-lite` → `gemini-3.5-flash-lite` → `gemini-2.5-flash-lite` → `gemini-flash-lite-latest` → `gemini-3.8-flash` → `gemini-3.5-flash` → `gemini-3.1-pro-preview`.

If any model returns HTTP `429 (Quota Exceeded)` or `404`, the system transparently retries with the next healthy model in milliseconds without interrupting the user experience.

5.2 3-Tier Hybrid Hindi Audio Streaming Architecture

Standard Web Speech API (`window.speechSynthesis`) silently fails on many Windows and budget Android browsers when asked to speak Devanagari Hindi (`hi-IN`) if the OS language pack is not installed. Sehat Sakha solves this permanently using a 3-tier architecture:

1. **Tier 1 — Vercel Serverless Audio Proxy (`/api/tts` in `api/tts.js`):** Streams natural Google Neural Hindi MP3 audio directly from the edge server with `Content-Type: audio/mpeg` and `Cache-Control: public, max-age=86400`, bypassing browser CORS/referrer blocks.
2. **Tier 2 — Direct Client Stream Fallback:** Falls back to direct client audio streaming if offline from the proxy.
3. **Tier 3 — Native Web Speech Synthesis Fallback:** Uses `SpeechSynthesisUtterance` with automatic voice discovery for offline environments.

5.3 Deterministic vs. Generative Separation of Concerns

A critical healthcare engineering principle in Sehat Sakha is separating **clinical classification** from **generative language**:

- **Generative AI (Gemini)** is used for *Optical Character Recognition (OCR)* and *empathetic plain-language translation*.
- **Deterministic TypeScript Code (`referenceEngine.ts`)** is used to compare the extracted number against the reference bounds and calculate chart positions. This guarantees zero mathematical hallucination—for example, a Hemoglobin of `11.2 g/dL` against a range of `13.5 - 17.5` will 100% deterministically be flagged as `low` (`सामान्य से कम`) and placed at the exact proportional coordinate on the 3-zone bar and Building Graph.

6. Presentation Script & Slide Deck Blueprint (For ChatGPT / Speaker Use)

When you upload this PDF to ChatGPT, you can ask it to expand or customize the following **7-Slide Presentation Structure & Speaker Script**:

SLIDE 1 Title & Hook — "Aapki Report, Aapki Bhasha Mein"

Visual: Sehat Sakha logo, Sakha mascot, and live URL (`ai-medical-simplifier.vercel.app`).
Speaker Script: *"Imagine a 65-year-old grandfather in a rural Indian village holding a blood test report. Every word is in technical English—Leukocytes, Creatinine, Bilirubin. He sees bold numbers, panics, and doesn't know what to ask his doctor. Today, we present **Sehat Sakha (सेहत सखा)**—an AI health companion that reads real medical reports and explains them in simple Hindi, both visually and through spoken voice."*

SLIDE 2 The 4 Barriers in Rural Healthcare Literacy

Visual: 4 icons showing Language Barrier, Complex Numbers, Low Vision/Literacy, and Short Doctor Visits.
Speaker Script: *"Existing health apps are built for urban, English-speaking tech users. Rural patients face four barriers: English medical jargon, confusing reference ranges, difficulty reading small mobile text, and not knowing what questions to ask their doctor."*

SLIDE 3 How Sehat Sakha Works — Real Multimodal OCR + Human Verification

Visual: Upload Screen → Gemini Vision OCR → Extraction Review Table.
Speaker Script: *"With Sehat Sakha, a user simply takes a photo or uploads a PDF of their lab report. Powered by Google Gemini Multimodal Vision, the app extracts every test name, value, unit, and reference range directly from the document—with zero fake or mock data. Crucially, we include a Human-in-the-Loop Verification step where the user can review and confirm the extracted numbers before proceeding."*

SLIDE 4 Mode A — "Simple by Default" & Native Hindi Voice Engine

Visual: Simple Explanation View + Active Green Hindi Audio Player with live subtitles.
Speaker Script: *"Our core design philosophy is 'Simple by Default, Detailed When Requested.' Mode A gives a clean, anxiety-free summary in plain Hindi. And for elderly users who struggle to read, one tap on 'Hindi Mein Sunein' streams natural Hindi speech via our serverless TTS engine—even translating cryptic units like 'mg/dL' into spoken 'मिलीग्राम प्रति डेसीलीटर' with live sentence-by-sentence subtitles."*

SLIDE 5 Mode B — Visual Metaphors Built for Everyone (Pie Chart & Building Graph)

Visual: Health Pie Chart, Building Bar Graph (इमारत ग्राफ), and 3-Zone Range Meter.
Speaker Script: "*When a user taps 'View Graphs', Mode B translates numbers into relatable visual metaphors. Our Health Pie Chart shows the exact ratio of normal vs. attention-needed tests. Our signature 'Imarat Graph' (Building Bar Graph) compares the patient's result pillar side-by-side against the normal reference building height. Every chart is paired with a simple Hindi text explanation and 1-tap audio.*"

SLIDE 6 Empowering the Doctor Visit & Automatic Health Vault

Visual: Ask Sakha Drawer + Doctor Questions Checklist + Sidebar "My Reports" with Date/Time tags.
Speaker Script: "*Sehat Sakha never diagnoses diseases or prescribes medicine. Instead, it generates a tailored checklist of questions for the patient's next doctor visit, offers the 'Ask Sakha' voice chat for follow-up doubts, and automatically archives every analyzed report in the sidebar 'My Reports' vault with exact date and time tags.*"

SLIDE 7 Hybrid Architecture — Combining Generative AI with Deterministic Safety

Visual: Architecture Diagram: React 19 + Gemini Vision OCR + Deterministic Reference Engine + Vercel Serverless `/api/tts`.
Speaker Script: "*Under the hood, we combine Generative AI with deterministic software rules. While Gemini handles OCR and empathetic translation across a 7-model fallback cascade, our deterministic Reference Engine mathematically verifies every value against clinical bounds so the AI never hallucinates a Normal or Abnormal status.*"

📋 Copy-Paste Prompt for ChatGPT (When You Upload This PDF)

```
"I have uploaded the detailed project report PDF for my web application 'Sehat Sakha (सेहत सखा) – Aapki Report, Aapki Bhasha Mein'. Please read the entire PDF carefully and: (1) Explain all its features and technical USPs clearly, (2) Write a compelling 5-minute presentation script with slide-by-slide speaker notes and stage directions for a live demo, and (3) Prepare 5 likely judge/panel questions with strong answers based on our hybrid AI + deterministic architecture."
```